

Intelligence as Agency

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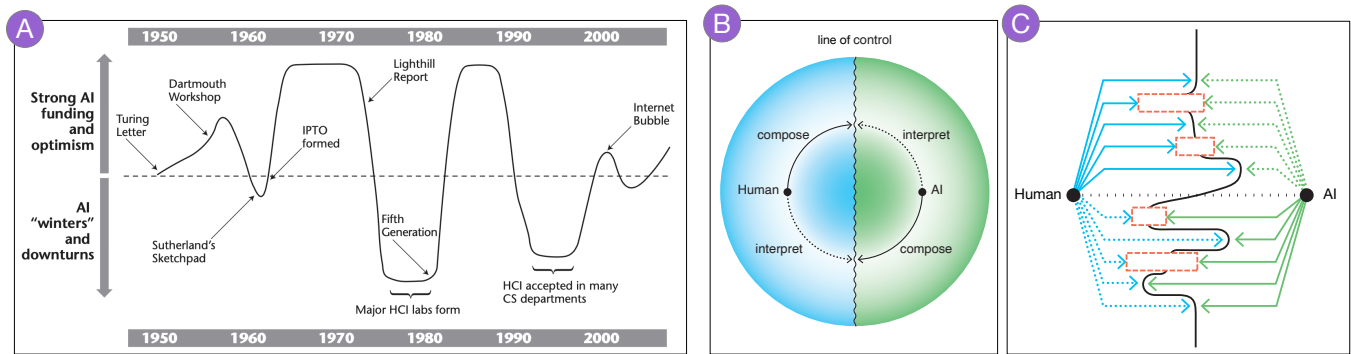


Figure 1: (a) A line chart reproduced from Jonathan Grudin’s 2009 AAAI article [10] depicting the competition for resources between AI and HCI; (b) A conceptual model of human-AI co-agency comprising *compositional* and *interpretive* agency; (c) Zooming into the line of control to visualize the range of tasks that might comprise an application, the different configurations of agency that might be required, and (in dotted orange regions) *open design spaces* for more open-ended human-AI interaction.

Abstract

In his 2009 AAAI article [10], Jonathan Grudin described AI and HCI as “two fields divided by a common focus”, noting how they “competed for intellectual and economic resources” (Fig. 1(a)). HCI is in its strongest position yet — with several senior HCI researchers leading human-centered AI teams, organizations, and institutes at major companies and universities. But, there continues to be the risk that history repeats itself: that HCI finds itself primarily reacting to advances in AI [19], rather than being a coequal discipline that exerts pressures that drive advances in AI as well. In this talk, I will propose two conceptual shifts that more explicitly center HCI values in an era of rapid progress in AI: (1) redefining intelligence as agency, the capacity to meaningfully act, rather than the capacity to perform a task; and (2) formulating design as the delegation of constrained agency, rather than solely the specification of affordances.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Interaction paradigms; Interaction design theory, concepts and paradigms; • **Computing methodologies** → Philosophical/theoretical foundations of artificial intelligence.

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Human-AI Interaction, Agency, AI, HCI

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1 Intelligence: From Competence to Agency

Dating back to Alan Turing’s foundational *Imitation Game* [26] — wherein a human observer was tasked with distinguishing whether natural language responses in a conversation were produced by a human or AI interlocutor — progress in AI has largely been measured in terms of how *competently* AI can *mimic human performance*. For instance, researchers at Google DeepMind have proposed a leveled matrix for tracking progress towards Artificial General Intelligence (AGI), where each level analogizes model performances to human performance (e.g., the second level is described as performance equivalent to “at least 50th percentile of skilled adults”) [20]. Similarly, AI ethics researchers have flagged that AGI may be no more than hype as today’s models are no more than “stochastic parrots” [5] that simply predict the next word, rather than learning the *meaning* or developing the higher-level *understanding* of textual content akin to how humans would [6]. While this paradigm has proven useful — providing metrics that afford benchmarking and empirical study — it is unable to grapple with the emergent interactional dynamics that generative AI has enabled — where context-sensitive, open-ended responses often surprise both users and model developers — nor how societal power structures may be reinforced and/or reconfigured.

In response, my talk will propose operationalizing intelligence in terms of *agency*. Agency has long-motivated discussions in AI and HCI [2, 3, 9, 11, 13, 17, 24] but has, perhaps surprisingly, been undertheorized in the constituent literatures. In a CHI 2023 paper [7], Bennett et al. observe that, in HCI, agency functions as an *umbrella construct*: a “broad concept or idea used loosely to encompass and account for a set of diverse phenomena.” Without a more precise definition of agency, they note that it has been difficult for the field to “accumulate and communicate knowledge” [7]. Thus, my talk will first offer a novel definition of agency drawn from research in linguistic anthropology [1, 8, 14]: *compositional* agency, or the capacity to express a goal-directed intent; and *interpretive* agency, or the ability to evaluate the efficacy of this expression. When individual people begin to engage in collaborative activity, they begin to blend their agencies such that their capacities for composition and interpretation are directed at one-another — that is, an agent composes actions to be interpreted by the other, who responds reciprocally. Through analogy, my talk will describe a conceptual model of human-AI interaction that recalls Licklider’s vision of “think[ing] in interaction with a computer in the same way that you think with a colleague whose competence supplements your own” [17] (an example of a particular fusion of human-AI co-agency is shown in Fig. 1(b)).

Critically, this model of agency is not normative — that is, it does not suggest an “optimal” blend of agency. Rather, as research in anthropology has shown, attitudes towards agency are shaped by history, culture, and politics [23, 25, 27]. Moreover, the specific fusion of agency that might be desirable is likely to also be dependent on the preferences of individual users as well as the specific tasks that are being jointly performed by the human-AI co-agent. Thus, in my talk, I will describe the opportunities for future analytic and empirical work this conceptual model of agency suggests. For instance, even with a 50-50 split of agency, for what tasks and under which cultures might either the human or AI opt to trade off one form of agency (i.e., composition or interpretation) for another? Or, should we always be alarmed if we track fused human-AI agency and notice a step-by-step reduction of human agency or might there be circumstances in which this may be desirable? Answering such questions will require developing novel approaches for measuring (and perhaps even benchmarking) agency, and my talk will sketch some potential directions that draw on cognitive science and behavioral economics.

2 Design: From Specification to Delegation

The second part of my talk will focus on what happens at the *line of control*: the region where the co-agents find common ground and negotiate their roles and responsibilities over the course of an interaction. Fig. 1(b) intentionally depicts the line of control as squiggly to indicate that it is not stable, but rather dynamic and shifting over time based — for a specific task, one agent might assume greater compositional agency and these roles may reverse for another task. By zooming into the line of control, as shown in Fig. 1(c), my talk will begin to unpack the role of interface designers.

Under a traditional design process, designers iteratively prototype and evaluate several candidate designs. Each design explores an alternate blend of co-agency (i.e., how much work does the user

need to perform, and what does the computer automatically “fill in”) and the set of designs considered is called a *design space*. A designer’s goal is typically to settle on a specific point in the design space — i.e., a fixed configuration of human-AI agency — and, as they repeat this process for each task in an application, the designer stitches together the line of control. This line of control does indeed wobble back-and-forth as each task in an application likely requires a difference balance of co-agency, but the *specific* way it wobbles is largely determined and fixed by the designer. As a result, except for a handful of instances of reappropriation [4, 18, 21], traditional user interfaces do not afford users the ability to assert alternate lines of control.

In my talk, I will describe how generative AI can facilitate a shift from *closed* design spaces — those that only a designer can explore — to more *open* ones (the dotted orange regions in Fig. 1(c)). In open design spaces, users can flexibly vary the agency they wish to assert and have the AI responsively modulate its agency to match. Early examples of such “open” interaction include tldraw’s Make Real feature — which allows users to modulate their compositional agency when authoring interactive web applications by mixing textual and graphical modalities as well as varying their level of precision — or Google Gemini’s disambiguation flow — which dynamically generates “bespoke” interfaces to facilitate user interpretation. Critically, open design spaces do not obviate the role of designers. Rather, they recall Latour’s notion of *delegation* [15]: a designer is distributing some of their power over constructing a *normative ground* [16] to the fused human-AI co-agent. Thus, the focus of a designer’s attention can now shift to questions of *how* and by *how much* should a design space be opened? By drawing on distributed cognition [12], my talk will call for a renewed focus on the *representations* that users and AI trade back-and-forth.

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